Software Requirements Specification

CAutomation / CCore / Pipeline Management

Document status: Baseline production contract for the Pipeline Management reference project. Only this SRS and the matching ATS are manually authored and approved at module level. This baseline hardens the module contract for deterministic planning and generation while preserving the contextual Pipeline plus reusable Task plus PipelineTask association model.

1. Purpose, Scope, and Source-of-Truth Rule

This SRS defines what the Pipeline Management epic must support from a business and product perspective. It is a functional contract, not an implementation design. It defines the approved hierarchy, functional concepts, workflows, rules, validation expectations, and acceptance criteria used by the CAutomation AI Engine.

- The SRS answers what must be built and why it is required.
- The ATS answers how the approved SRS is implemented.
- The ATS must not introduce business requirements absent from this SRS.
- If generated artifacts conflict with this SRS, this SRS wins.
- If behavior must change, this SRS is updated and approved before regeneration.

2. Business Context and Approved Hierarchy

The Pipeline Management epic belongs to the following approved CAutomation hierarchy. The labels must appear consistently in generated agile artifacts, UI labels, reports, and traceability output.

Level	Approved value	Business meaning
Project	CAutomation	The top-level software initiative and product family boundary.
Product	CCore	The core CAutomation product that owns shared platform capabilities.
Module	Pipeline Management	The automation module that manages AI-driven generation workflows.
Epic	Pipeline Management	The functional epic for defining reusable tasks, creating contextual pipelines, executing pipeline runs, and reporting results.

3. In Scope and Out of Scope

In scope	Out of scope
Maintain Projects, Products, Modules, Pipelines, reusable Tasks, PipelineTask associations, Pipeline Executions, Pipeline Task Executions, and Reports.	Changing repository folder structure or creating new AI Engine pipeline folders.
Maintain metadata for Project Type, Project Status, Project Target, Product Type, Product Status, Product Target, Module Type, Module Status, and Module Target.	Defining a platform-wide authentication, authorization, or tenant model.
Create pipelines inside a selected Project/Product/Module context and add reusable tasks to those pipelines.	Executing arbitrary AI prompts outside approved pipeline definitions.
Generate agile planning artifacts from approved functional scope.	Inventing requirements during planning, generation, validation, apply, or verification.
Expose deterministic validation and reporting behavior.	Replacing project-level contracts; project-level SRS/ATS documents are deferred to a later iteration.

4. Actors and Permissions

Actor	Business responsibility	Required permissions
Product Owner	Approves SRS scope, acceptance criteria, and business rule changes.	Read all, approve specification changes.
Automation Architect	Maintains hierarchy, metadata, reusable task catalog, pipeline definitions, task placement, and lifecycle.	Create, read, update, archive, restore, validate.
Developer	Consumes generated artifacts and reviews implementation output.	Read, validate, view reports.
Reviewer	Inspects generated plans, validation findings, execution reports, and traceability.	Read, view reports.
AI Engine	Consumes approved SRS and ATS to generate artifacts.	Read approved contracts only.

5. Business Domain Model

The domain separates the contextual pipeline definition from the reusable task catalog. A Pipeline is created for a selected Project/Product/Module context. A Task is reusable and may appear in many Pipelines. PipelineTask is the association that places one reusable Task inside one Pipeline with order, requirement, configuration, retry policy, and failure behavior. This prevents the system from treating a task as owned by only one pipeline.

Entity	Business description	Parent or association	Lifecycle expectation
Project	Top-level initiative such as CAutomation.	None	Draft, Active, Suspended, Archived.
Project Type/Status/Target	Metadata that classifies project purpose, lifecycle, and delivery target.	None	Active or Archived lookup value.
Product	Product owned by a project, such as CCore.	Project	Draft, Active, Suspended, Archived.
Product Type/Status/Target	Metadata that classifies product purpose, lifecycle, and delivery target.	None	Active or Archived lookup value.
Module	Module owned by a product, such as Pipeline Management.	Product	Draft, Active, Suspended, Archived.
Module Type/Status/Target	Metadata that classifies module purpose, lifecycle, and delivery target.	None	Active or Archived lookup value.
Pipeline	Named, versioned workflow definition created inside selected Project/Product/Module context.	Project + Product + Module context	Draft, Active, Deprecated, Archived.
Task	Reusable task definition that can be used in many pipelines.	Global reusable catalog	Draft, Active, Disabled, Archived.
PipelineTask	Association that places a Task into a Pipeline and stores sequence/order/configuration.	Pipeline plus Task	Active, Disabled, Archived.
Pipeline Execution	Run record created when a pipeline is executed.	Pipeline	Pending, Running, Passed, Failed, Cancelled.
Pipeline Task	Run record created	Pipeline Execution	Pending, Running,

Entity	Business description	Parent or association	Lifecycle expectation
Execution	for one PipelineTask during a pipeline execution.	plus PipelineTask	Passed, Failed, Skipped.
Report	Generated report associated with a pipeline execution or task execution.	Execution record	Generated, Reviewed, Archived.

6. Functional Requirements

ID	Requirement	Area
SRS-FR-001	The system shall maintain Project records with code, name, description, type, status, target, ownership, and audit information.	Project
SRS-FR-002	The system shall maintain Product records under exactly one Project.	Product
SRS-FR-003	The system shall maintain Module records under exactly one Product.	Module
SRS-FR-004	The system shall maintain lookup metadata for type, status, and target for Projects, Products, and Modules.	Metadata
SRS-FR-005	The system shall create Pipeline definitions after the user selects Project, Product, and Module context.	Pipeline
SRS-FR-006	The system shall maintain a reusable Task catalog independent of any single Pipeline.	Task
SRS-FR-007	The system shall allow one Task to be used by multiple Pipelines through PipelineTask association records.	PipelineTask
SRS-FR-008	The system shall store PipelineTask ordering, required/optional behavior,	PipelineTask

ID	Requirement	Area
	configuration, retry policy, and failure behavior per Pipeline.	
SRS-FR-009	The system shall prevent activation of a Pipeline unless required context, metadata, and at least one active PipelineTask exist.	Lifecycle
SRS-FR-010	The system shall create Pipeline Execution records that snapshot the selected Pipeline version and ordered PipelineTask list.	Execution
SRS-FR-011	The system shall create Pipeline Task Execution records for each executable PipelineTask in the execution snapshot.	Execution
SRS-FR-012	The system shall expose execution status, task status, timestamps, messages, and report references.	Reporting
SRS-FR-013	The system shall generate planning artifacts from SRS functional requirements, rules, workflows, and acceptance criteria.	Planning
SRS-FR-014	The system shall never allow downstream AI stages to invent business scope not present in approved SRS.	AI Boundary

7. Business Data and Field Rules

Entity scope	Business field	Rule
All primary entities	id	Required stable identifier. User-visible only where useful for traceability.
All primary entities	name	Required human-readable name. Must be unique within approved parent boundary where applicable.
All primary entities	code	Required stable short code

Entity scope	Business field	Rule
		for generation, traceability, and reporting.
All primary entities	description	Optional business description.
Project/Product/Module	type/status/target	Required references to active metadata values before activation.
Pipeline	project/product/module context	Required context selected before pipeline creation; product must belong to project and module must belong to product.
Pipeline	version	Required version value. Active pipelines are versioned.
Task	task_key	Required stable reusable task key, unique in task catalog.
PipelineTask	pipeline + task	Required association between one Pipeline and one reusable Task.
PipelineTask	sequence_order	Required positive integer unique within the pipeline.
PipelineTask	configuration/retry/failure behavior	Optional or required according to task type; stored per pipeline usage, not globally on the reusable task.
Execution records	status	Required lifecycle status. Must follow allowed transitions.
Report	report_type	Required classifier such as planning, validation, apply, verification, or execution summary.

8. Relationship and Lifecycle Rules

- A Product cannot exist without a Project.
- A Module cannot exist without a Product.
- A Pipeline is created only after selecting a valid Project, Product, and Module context.
- A Pipeline context must be internally consistent: Product belongs to Project, and Module belongs to Product.
- A Task is reusable and must not be owned by one Pipeline, Project, Product, or Module.
- A PipelineTask must reference exactly one Pipeline and exactly one Task.

- The same Task may be used by multiple Pipelines through multiple PipelineTask records.
- A PipelineTask sequence order must be unique within its Pipeline.
- A Pipeline Execution must reference the Pipeline version and ordered PipelineTask snapshot executed.
- A Pipeline Task Execution must reference the Pipeline Execution and the PipelineTask snapshot or identity.
- Archived parent records must not accept new active child records.
- Deletion is not a normal business operation; archive is the expected lifecycle action.

9. User Workflows

Workflow	Required behavior
Create hierarchy	User creates or selects Project, Product, and Module before defining pipelines.
Configure metadata	User selects valid type, status, and target metadata for Project, Product, and Module.
Define reusable task	User creates or selects a reusable Task with stable key, name, type, description, and default responsibility.
Define pipeline	User selects Project, Product, and Module context, then creates a named and versioned Pipeline.
Add tasks to pipeline	User adds reusable Tasks to a Pipeline through PipelineTask records with sequence order and usage-specific configuration.
Validate pipeline	User validates completeness before activation. Validation returns stable rule identifiers.
Execute pipeline	System creates execution records from the approved pipeline and ordered PipelineTask snapshot.
Review reports	User reviews generated reports linked to execution records and task execution records.

10. Validation Rules

Rule ID	Rule
SRS-VAL-001	Required fields must be present before save when the field is mandatory for the current action.
SRS-VAL-002	Names and codes must be unique within the approved parent boundary.
SRS-VAL-003	A status transition must be one of the

Rule ID	Rule
	allowed lifecycle transitions.
SRS-VAL-004	Activation requires required metadata references to be active.
SRS-VAL-005	Pipeline activation requires selected Project/Product/Module context to be valid and active.
SRS-VAL-006	Pipeline activation requires at least one active PipelineTask.
SRS-VAL-007	PipelineTask sequence_order must be unique and positive within the Pipeline.
SRS-VAL-008	A PipelineTask must reference an active reusable Task before activation.
SRS-VAL-009	Execution records must not be edited as definitions after creation.
SRS-VAL-010	Reports must reference an existing execution record.
SRS-VAL-011	Runtime execution parameter merge evaluations must perform a shallow merge of execution_parameters_json into the immutable baseline pipeline_task.configuration_json snapshot. Duplicate keys provided at execution runtime must override matching base configuration keys. The resulting merged configuration snapshot shall remain completely immutable for the lifetime of that specific Pipeline Execution.

11. Acceptance Criteria

ID	Acceptance criterion
SRS-AC-001	Generated agile artifacts include Project, Product, Module, Pipeline, reusable Task, PipelineTask, Execution, and Reporting scope.
SRS-AC-002	Generated schema includes metadata lookup tables for type, status, and target at Project, Product, and Module levels.
SRS-AC-003	Generated schema represents Pipeline and Task as separate entities connected by PipelineTask many-to-many association records.
SRS-AC-004	Generated APIs preserve selected

ID	Acceptance criterion
	Project/Product/Module pipeline context and do not expose child records outside their approved boundary.
SRS-AC-005	Generated UI supports create, view, update, validate, archive, restore, task selection, pipeline task ordering, execution, and report review where applicable.
SRS-AC-006	Generated tests assert validation rule identifiers, lifecycle transitions, many-to-many task reuse, and execution snapshot behavior.
SRS-AC-007	Generated reports trace execution output back to approved pipeline, PipelineTask, and reusable Task definitions.

12. Deterministic Generation Boundary

Planning may derive epics, features, user stories, tasks, and acceptance tests only from the approved SRS. Generation may consume both SRS and ATS. If required information is missing, the AI Engine must produce a specification gap and must not infer hidden business behavior. The current iteration does not add new AI Engine pipeline folders and does not create project-level contracts; those project-level documents are deferred.

13. Entity-Level Functional Contract

Entity	Create	Read	Update	Special functional rule
Project	Authorized architect may create.	List and detail views required.	Allowed while not archived.	Represents CAutomation or future top-level initiatives.
Product	Created under one Project.	List under Project and detail required.	Allowed while parent is usable.	Represents CCore in current epic.
Module	Created under one Product.	List under Product and detail required.	Allowed while parent is usable.	Represents Pipeline Management in current epic.
Pipeline	Created after selecting Project/Product /Module context.	List by context and detail required.	Allowed while draft or inactive.	Defines a versioned contextual workflow.
Task	Created in	List/search for	Allowed while	Reusable across

Entity	Create	Read	Update	Special functional rule
	reusable task catalog.	selection into pipelines.	not locked by immutable execution snapshots.	multiple pipelines.
PipelineTask	Created by adding a Task to a Pipeline.	Shown in ordered sequence.	Allowed while pipeline is editable.	Represents usage/configuration of a reusable Task in one Pipeline.
Pipeline Execution	Created by execution action.	Detail and status view required.	Status updates only.	Represents one run of a pipeline version and task snapshot.
Pipeline Task Execution	Created from pipeline execution.	Shown under execution detail.	Status/message updates only.	Represents one PipelineTask run.
Report	Generated from planning, validation, execution, apply, or verification.	List and detail views required.	Review status may change.	Provides traceability and evidence.

14. Lifecycle Transition Contract

Record type	Allowed normal transitions	Blocked transitions
Project/Product/Module	Draft to Active, Active to Suspended, Suspended to Active, Active/Suspended to Archived, Archived to Draft or Suspended when restored.	Archived directly to Active without validation; Active to Draft after child records exist.
Pipeline	Draft to Active, Active to Deprecated, Deprecated to Archived, Draft to Archived, Archived to Draft when restored.	Active without valid context, active metadata, and at least one active PipelineTask.
Task	Draft to Active, Active to Disabled, Disabled to Active, any non-running state to Archived.	Archived or disabled task selected for new active PipelineTask without reactivation.
PipelineTask	Active to Disabled, Disabled to Active, Active/Disabled to Archived.	Duplicate sequence order in the same Pipeline; missing reusable Task.

Record type	Allowed normal transitions	Blocked transitions
Pipeline Execution	Pending to Running, Running to Passed, Running to Failed, Pending/Running to Cancelled.	Passed or Failed back to Running; Cancelled to Passed.
Pipeline Task Execution	Pending to Running, Pending to Skipped, Running to Passed, Running to Failed.	Passed/Failed back to Pending; Skipped to Running after parent execution is complete.
Report	Generated to Reviewed, Generated/Reviewed to Archived.	Archived to Reviewed without restore action.

Asynchronous Execution Orchestration Mandate: The backend application layer must utilize native FastAPI BackgroundTasks as the mandatory backend execution orchestration mechanism. The native BackgroundTasks engine shall be authoritatively responsible for managing and driving all Pipeline Execution and Pipeline Task Execution lifecycle state transitions. This engine must safely process and execute the sequential asynchronous workflow loops, moving active execution statuses strictly through the approved state machine path: Pending -> Running -> Passed / Failed / Cancelled. Implementing or incorporating alternative external execution frameworks, distributed message brokers, or parallel task consumer loops is strictly prohibited and remains entirely outside the scope of this module contract unless the Project Engineering Contract is formally revised.

15. Agile Artifact Generation Contract

Generated artifact	SRS source material	Required deterministic output
Epic	Business hierarchy and SRS purpose/scope.	One epic named Pipeline Management under CAutomation / CCore.
Features	Functional requirements and business domain model.	Feature groups for hierarchy management, metadata management, reusable task catalog, pipeline definition, execution, reporting, and validation.
User stories	Actors, workflows, functional requirements, and validation rules.	Stories with role, goal, reason, acceptance criteria, and traceability IDs.
Implementation tasks	SRS functional areas plus ATS technical mapping.	Tasks for schema, API, service, frontend, validation, tests, and reports.
Acceptance tests	SRS acceptance criteria and	Positive and negative tests

Generated artifact	SRS source material	Required deterministic output
	validation rules.	using stable rule identifiers.
Specification gaps	Missing required SRS or ATS details.	Explicit blocking gap records instead of invented behavior.

16. Business Scenarios

Scenario	Given	When	Then
Create CAutomation hierarchy	No current hierarchy exists for the epic.	An authorized architect creates CAutomation, CCore, and Pipeline Management with active metadata.	The hierarchy is available for pipeline definition and appears in generated planning artifacts.
Create reusable task	The task catalog is available.	The architect creates a task such as Validate Output or Generate Report.	The task can be selected into multiple pipelines without duplication.
Create contextual pipeline	A Project/Product/Module context exists and is active.	The architect selects CAutomation, CCore, and Pipeline Management and creates a pipeline.	The pipeline is saved as Draft in the selected context.
Add reusable tasks	A draft pipeline exists and reusable tasks exist.	The architect adds tasks to the pipeline with sequence order and configuration.	PipelineTask records define the ordered workflow without changing the reusable task definitions.
Validate invalid pipeline	A pipeline has missing active tasks or inactive metadata.	The user validates or attempts activation.	The system returns stable validation rule identifiers and blocks activation.
Execute active pipeline	A pipeline is active and contains active ordered PipelineTasks.	The user starts execution.	A pipeline execution and task execution records are created from the snapshot.
Review reports	An execution has completed or failed.	The user opens execution reports.	The report list shows generated summaries and links them to the execution and task statuses.

17. Generation Readiness Pass

This section defines the functional completeness checks that must pass before the AI Engine treats the module SRS and ATS as generation-ready. The checks do not add new business scope; they prevent generation from filling gaps with invented requirements.

Readiness area	Required SRS decision	Generation behavior if missing
Business hierarchy	Project CAutomation, Product CCore, and Module Pipeline Management must be present.	Block generation and report missing hierarchy contract.
Pipeline context	Pipeline must be created only after Project, Product, and Module are selected and validated.	Block pipeline generation and report missing context rule.
Task reuse	Task must be reusable globally; PipelineTask must associate one Task with one Pipeline.	Block schema and API generation if Pipeline and Task are collapsed into one table.
Lifecycle	Allowed statuses and transitions must be explicit for hierarchy records, pipelines, tasks, executions, and reports.	Generate lifecycle gap instead of inventing statuses.
Validation	Validation rule IDs must be stable and traceable.	Generate validation-gap report if rule identifiers are missing.
Acceptance evidence	Each generated story, test, API, or table must cite at least one SRS requirement or acceptance criterion.	Reject untraced output during validation.

18. Approved Functional Status Values

The following status value sets are approved at business level. The ATS may implement them as lookup rows or constrained constants, but generated artifacts must preserve the exact semantic values.

Record type	Approved statuses	Business note
Project	Draft, Active, Suspended, Archived	Archived is not deleted and must remain reportable.
Product	Draft, Active, Suspended, Archived	Product status is evaluated before module and pipeline activation.

Record type	Approved statuses	Business note
Module	Draft, Active, Suspended, Archived	Module must be Active before creating an Active pipeline.
Pipeline	Draft, Active, Deprecated, Archived	Deprecated pipelines are retained for traceability but not preferred for new executions.
Task	Draft, Active, Disabled, Archived	Disabled tasks cannot be selected into newly activated pipelines.
PipelineTask	Active, Disabled, Archived	Disabled associations are excluded from new execution snapshots.
PipelineExecution	Pending, Running, Passed, Failed, Cancelled	Passed, Failed, and Cancelled are terminal.
PipelineTaskExecution	Pending, Running, Passed, Failed, Skipped	Skipped applies when optional or dependency-blocked tasks are not run.
Report	Generated, Reviewed, Archived	Report records must remain linked to their execution source.

19. Specification Gap Handling

A specification gap is a blocking finding that identifies missing information needed for deterministic generation. The AI Engine must report gaps rather than invent missing behavior, fields, endpoints, screens, tests, or repository paths.

Gap type	Example	Required output
Business gap	A required workflow is referenced but has no acceptance criterion.	Gap record with SRS section, missing decision, severity, and recommended clarification.
Data gap	An entity appears in the SRS but lacks required business field rules.	Gap record naming the entity and missing field rule.
Lifecycle gap	A status exists without allowed transition rules.	Gap record naming record type and missing transition.
Traceability gap	Generated artifact has no SRS requirement or acceptance criterion source.	Reject generated artifact and report traceability failure.

Gap type	Example	Required output
Implementation mapping gap	SRS requirement has no ATS implementation mapping.	Block generation until ATS maps the SRS requirement.

20. Minimum Generation Scope

A generation run for this module is complete only when it can deterministically produce agile artifacts, database schema, backend APIs and services, frontend screens, validation rules, tests, and reports for the approved Pipeline Management scope. Project-level SRS and ATS contracts remain deferred and must not be generated in this iteration.